

DEGREE-SIXTEEN COHOMOLOGY OF MODULI SPACES OF STABLE CURVES IN GENUS AT MOST SEVEN IS TATE

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ABSTRACT. We prove that, for every stable pair (g, n) with $g \leq 7$, the rational Hodge structure $H^{16}(\overline{\mathcal{M}}_{g,n}; \mathbf{Q})$ is a direct sum of copies of $\mathbf{Q}(-8)$. The proof extends the boundary induction of Canning–Larson–Payne by one even degree. Its new open inputs are obtained from their primitive-quotient formula, virtual cohomological dimension, and the Ionel–Looijenga vanishing theorem for tautological classes. The open Borel–Moore term vanishes whenever $3 \leq g \leq 7$ and $2g + n \geq 11$. In particular, the lowest-weight contribution from $\mathcal{M}_{6,6}$ is zero, so boundary homology surjects onto $H_{16}(\overline{\mathcal{M}}_{6,6}; \mathbf{Q})$. We also show that the unrestricted degree-sixteen problem reduces to two open inputs, on $\mathcal{M}_8$ and $\mathcal{M}_{8,1}$.

1. INTRODUCTION

Let $\overline{\mathcal{M}}_{g,n}$ be the smooth proper Deligne–Mumford stack of stable complex curves of genus g with n ordered marked points, and let $\mathcal{M}_{g,n} \subset \overline{\mathcal{M}}_{g,n}$ be the open substack of smooth pointed curves. A pair (g, n) is called stable if $2g - 2 + n > 0$. All cohomology and homology groups in this paper have rational coefficients and carry their natural mixed Hodge structures.

Canning, Larson, and Payne proved that the rational cohomology of $\overline{\mathcal{M}}_{g,n}$ is Tate in every even degree at most fourteen, uniformly in g and n [6, Theorem 1.5(3) and the discussion following it]. Degree sixteen is the first even degree not covered by this result. Its status in the general Hodge- and Tate-theoretic predictions is discussed by Payne [15, Section 4]. We prove the prediction through genus seven.

Theorem 1.1. *Let (g, n) be stable and suppose that $g \leq 7$. If*

$$b_{g,n} := \dim_{\mathbf{Q}} H^{16}(\overline{\mathcal{M}}_{g,n}; \mathbf{Q}),$$

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then there is an isomorphism of polarizable rational Hodge structures

$$H^{16}(\overline{\mathcal{M}}_{g,n}; \mathbf{Q}) \cong \mathbf{Q}(-8)^{\oplus b_{g,n}}. \quad (1)$$

The assertion is about Hodge type. It neither determines $b_{g,n}$ nor asserts that the group in (1) is generated by algebraic or tautological cycle classes. This distinction is essential: Tate Hodge type alone does not imply the Hodge conjecture.

The case that motivated the argument has an additional conclusion. Write $\widetilde{\partial\mathcal{M}}_{g,n}$ for the normalization of the reduced boundary $\overline{\mathcal{M}}_{g,n} \setminus \mathcal{M}_{g,n}$.

Corollary 1.2. *The proper pushforward*

$$H_{16}(\widetilde{\partial\mathcal{M}}_{6,6}; \mathbf{Q}) \longrightarrow H_{16}(\overline{\mathcal{M}}_{6,6}; \mathbf{Q}) \quad (2)$$

is surjective. In particular, if $b_{16} = \dim_{\mathbf{Q}} H^{16}(\overline{\mathcal{M}}_{6,6}; \mathbf{Q})$, then

$$H^{16}(\overline{\mathcal{M}}_{6,6}; \mathbf{Q}) \cong \mathbf{Q}(-8)^{\oplus b_{16}}.$$

The surjectivity follows from a vanishing on the open moduli stack:

$$W_{26}H^{26}(\mathcal{M}_{6,6}; \mathbf{Q}) = 0, \quad W_{-16}H_{16}^{\text{BM}}(\mathcal{M}_{6,6}; \mathbf{Q}) = 0. \quad (3)$$

More generally, the critical open groups that occur at the boundary of the known Chow–Künneth ranges all vanish:

$$\begin{aligned} W_{20}H^{20}(\mathcal{M}_{3,12}; \mathbf{Q}) &= 0, \\ W_{22}H^{22}(\mathcal{M}_{4,10}; \mathbf{Q}) &= 0, \\ W_{24}H^{24}(\mathcal{M}_{5,8}; \mathbf{Q}) &= 0, \\ W_{26}H^{26}(\mathcal{M}_{6,6}; \mathbf{Q}) &= 0, \\ W_{28}H^{28}(\mathcal{M}_{7,4}; \mathbf{Q}) &= 0. \end{aligned} \quad (4)$$

The second row lies in the published Chow–Künneth range. The other four rows, including the genus-five case, follow from the one-marking argument in Section 3.

The proof of Theorem 1.1 follows the induction in [6, Section 6.2]. The localization sequence separates degree-sixteen homology into a normalized-boundary image and a lowest-weight Borel–Moore quotient on $\mathcal{M}_{g,n}$. Bergström–Faber–Payne vanishing makes the open quotient zero outside a finite range. Within that range, the published Chow–Künneth generation results control every case except (3, 12), (5, 8), (6, 6), and (7, 4). Canning–Larson–Payne’s primitive-quotient formula removes one marking in these cases. Virtual cohomological dimension kills both the primitive local-system quotient and the unmultiplied pullback term. The remaining cotangent-line term comes from a tautological group at the inclusive endpoint of their published range, and Ionel’s theorem makes that source zero.

On the boundary, every component of the normalization is a finite quotient of either one smaller moduli stack or a product of two. Odd–odd Künneth summands in total degree sixteen vanish because one odd degree is at most seven. Positive even–even summands are Tate by the uniform result through degree fourteen, and the two axis summands follow by induction. Rational averaging handles graph automorphisms, while semisimplicity of polarizable pure Hodge structures splits the resulting same-weight extensions.

The same method isolates the remaining obstruction to the unrestricted degree-sixteen problem.

Theorem 1.3. *Suppose that*

$$W_{-16}H_{16}^{\text{BM}}(\mathcal{M}_{8,n}; \mathbf{Q})$$

is a direct sum of copies of $\mathbf{Q}(8)$ for $n = 0, 1$. Then $H^{16}(\overline{\mathcal{M}}_{g,n}; \mathbf{Q})$ is a direct sum of copies of $\mathbf{Q}(-8)$ for every stable pair (g, n) .

Thus the genus-at-most-seven theorem is not an isolated collection of examples: it leaves precisely two genus-eight open inputs. The apparent third endpoint, $\mathcal{M}_{8,2}$, vanishes by Wong’s theorem [18, Theorem 4.1]. We return to this reduction in Section 6.

2. MIXED HODGE STRUCTURES AND THE BOUNDARY SEQUENCE

2.1. Weights, twists, and duality. The Tate Hodge structure $\mathbf{Q}(m)$ has weight $-2m$ and Hodge type $(-m, -m)$. Thus $\mathbf{Q}(-8)$ has weight 16 and type $(8, 8)$, whereas $\mathbf{Q}(8)$ has weight -16 and type $(-8, -8)$. We use the increasing weight filtration $W_{\bullet}$. The mixed Hodge theory used below is due to Deligne [7, 8]; for cohomology and duality on Deligne–Mumford stacks, see Behrend [1]. Throughout, the rational cohomology of a Deligne–Mumford stack of finite type over $\mathbf{C}$ agrees, with its mixed Hodge structure, with that of its coarse moduli space; for the stacks considered here the coarse space is a quasi-projective variety with finite quotient singularities, so rational Poincaré duality holds in the smooth case.

If X is smooth of pure complex dimension d , Poincaré duality gives an isomorphism of mixed Hodge structures

$$H_i^{\text{BM}}(X; \mathbf{Q}) \cong H^{2d-i}(X; \mathbf{Q})(d). \quad (5)$$

If X is proper, ordinary homology is the Hodge dual of ordinary cohomology:

$$H_i(X; \mathbf{Q}) \cong H^i(X; \mathbf{Q})^{\vee}. \quad (6)$$

For smooth proper X , these structures are polarizable and pure of weights i in cohomology and $-i$ in homology.

Every subobject or quotient of $\mathbf{Q}(m)^{\oplus N}$ in the category of rational mixed Hodge structures is a direct sum of copies of $\mathbf{Q}(m)$. Moreover, polarizable pure rational Hodge structures of fixed weight form a semisimple category. Consequently, an extension of copies of one Tate structure by copies of the same Tate structure splits noncanonically.

2.2. Normalization of the boundary. Knudsen’s clutching morphisms [12] describe the normalized boundary by one-edge stable graphs. Here a stable graph of type (g, n) is a connected graph with legs labelled by $\{1, \dots, n\}$ and with a nonnegative integer g_v at each vertex, such that

$$g = h^1(\Gamma) + \sum_v g_v, \quad 2g_v - 2 + n_v > 0$$

for every vertex; n_v counts incident half-edges and legs. Its automorphisms preserve every leg label and every vertex genus. In particular, the automorphism of a loop may exchange its two branches. If Γ has one edge, put

$$\overline{\mathcal{M}}_\Gamma := \prod_{v \in V(\Gamma)} \overline{\mathcal{M}}_{g_v, n_v}.$$

Then

$$\widetilde{\partial \mathcal{M}}_{g,n} \cong \coprod_{\substack{[\Gamma] \text{ stable of type } (g,n) \\ |E(\Gamma)|=1}} [\overline{\mathcal{M}}_\Gamma / \text{Aut}(\Gamma)]. \quad (7)$$

This is the description used in [6, Section 2.1]. A loop gives the nonseparating component $[\overline{\mathcal{M}}_{g-1, n+2}/S_2]$; an edge joining two vertices gives a finite quotient of a product.

For a finite group G acting algebraically on a complex Deligne–Mumford stack Y , the Cartan–Leray spectral sequence and rational averaging give

$$H^*([Y/G]; \mathbf{Q}) \cong H^*(Y; \mathbf{Q})^G \quad (8)$$

as mixed Hodge structures. Indeed, $H^p(G, V) = 0$ for $p > 0$ over $\mathbf{Q}$. Thus finite graph-automorphism quotients do not introduce new Hodge types.

2.3. The right-exact sequence. Let

$$\nu: \widetilde{\partial \mathcal{M}}_{g,n} \longrightarrow \overline{\mathcal{M}}_{g,n}$$

be the natural finite map.

Proposition 2.1 (Canning–Larson–Payne). *For every stable pair (g, n) and every integer i , there is a right-exact sequence of mixed Hodge structures*

$$H_i(\widetilde{\partial \mathcal{M}}_{g,n}; \mathbf{Q}) \xrightarrow{\nu_*} H_i(\overline{\mathcal{M}}_{g,n}; \mathbf{Q}) \longrightarrow W_{-i} H_i^{\text{BM}}(\mathcal{M}_{g,n}; \mathbf{Q}) \longrightarrow 0. \quad (9)$$

This is precisely Equation (6.1) of [6]. We use that published mixed-Hodge-theoretic statement directly; in particular, the first map in (9) has neither a degree shift nor a Tate twist.

For $i = 16$, let $I_{g,n}$ be the image of the first map. We obtain a short exact sequence

$$0 \longrightarrow I_{g,n} \longrightarrow H_{16}(\overline{\mathcal{M}}_{g,n}; \mathbf{Q}) \longrightarrow W_{-16}H_{16}^{\text{BM}}(\mathcal{M}_{g,n}; \mathbf{Q}) \longrightarrow 0. \quad (10)$$

3. THE OPEN CONTRIBUTION IN GENUS AT MOST SEVEN

3.1. The primitive quotient. For $1 \leq i \leq n$, let

$$\pi_i: \mathcal{M}_{g,n} \longrightarrow \mathcal{M}_{g,n-1}$$

forget the i th marking, and let $\psi_i \in H^2(\mathcal{M}_{g,n}; \mathbf{Q})$ be the first Chern class of its cotangent line. Following [6, Sections 2 and 3], define

$$\Phi_{g,n}^k := \sum_{i=1}^n \pi_i^* W_k H^k(\mathcal{M}_{g,n-1}; \mathbf{Q}), \quad (11)$$

$$\Psi_{g,n}^k := \sum_{i=1}^n \psi_i \smile \Phi_{g,n}^{k-2}. \quad (12)$$

Both are subspaces of $W_k H^k(\mathcal{M}_{g,n}; \mathbf{Q})$. Let $\lambda = (\lambda_1 \geq \dots \geq \lambda_g \geq 0)$ be a partition with $|\lambda| := \sum_i \lambda_i = n$. Write $\mathbb{V}_\lambda$ for the associated irreducible symplectic local system on $\mathcal{M}_g$, and write V_{λ^τ} for the irreducible S_n -representation corresponding to the transpose partition. These are the conventions of [6, Section 3].

Proposition 3.1 (Canning–Larson–Payne). *If $g \geq 2$, there is an isomorphism of $\mathbf{Q}$ -vector spaces*

$$\frac{W_k H^k(\mathcal{M}_{g,n}; \mathbf{Q})}{\Phi_{g,n}^k + \Psi_{g,n}^k} \cong \bigoplus_{|\lambda|=n} W_k H^{k-n}(\mathcal{M}_g; \mathbb{V}_\lambda) \otimes V_{\lambda^\tau}. \quad (13)$$

Proof. This is [6, Lemma 3.1(a)]. In its proof, the summands of relative cohomological degree two are identified by the relative projector π_2 of Petersen–Tavakol–Yin [17, Sections 5.1 and 5.2.2]. $\square$

We use only the following consequence: if every summand on the right of (13) vanishes, then $W_k H^k(\mathcal{M}_{g,n}; \mathbf{Q}) = \Phi_{g,n}^k + \Psi_{g,n}^k$, so the vanishing of $\Phi_{g,n}^k$ and of $\Psi_{g,n}^k$ forces $W_k H^k(\mathcal{M}_{g,n}; \mathbf{Q}) = 0$. No compatibility of (13) with Hodge structures is needed.

The following coefficient form of virtual-cohomological-dimension vanishing will be used repeatedly.

Lemma 3.2. *Let Γ be virtually torsion-free of virtual cohomological dimension d , and let L be a finite-dimensional rational Γ -module. Then*

$$H^q(\Gamma; L) = 0 \quad (q > d).$$

Proof. Choose a torsion-free normal subgroup $\Gamma' \triangleleft \Gamma$ of finite index; by Serre's theorem it has cohomological dimension d . Then $H^q(\Gamma'; L) = 0$ for $q > d$. In the Hochschild–Serre spectral sequence for Γ' in Γ , the quotient is finite. Its higher rational cohomology vanishes, so

$$H^q(\Gamma; L) \cong H^q(\Gamma'; L)^{\Gamma/\Gamma'} = 0 \quad (q > d).$$

See [3, Chapters VII and VIII]. $\square$

Teichmüller space is contractible, so the relevant stack cohomology with local coefficients agrees with mapping-class-group cohomology. Harer proved that, for $g \geq 2$ and $n > 0$,

$$\mathrm{vcd}(\mathrm{Mod}_g) = 4g - 5, \quad \mathrm{vcd}(\mathrm{PMod}_{g,n}) = 4g - 4 + n \quad (14)$$

[9, Theorem 4.1].

3.2. Chow–Künneth generation and vanishing. A Chow group in this paper always has rational coefficients: $A^*(X) = A^*(X)_{\mathbf{Q}}$. Let $R^*(\overline{\mathcal{M}}_{g,n}) \subset A^*(\overline{\mathcal{M}}_{g,n})$ be the tautological ring, and let $R^*(\mathcal{M}_{g,n}) \subset A^*(\mathcal{M}_{g,n})$ be its image under restriction; since all boundary generators restrict to zero, the ring $R^*(\mathcal{M}_{g,n})$ is generated by the ψ - and κ -classes. Define tautological cohomology by

$$\begin{aligned} \mathrm{RH}^{2c}(\mathcal{M}_{g,n}; \mathbf{Q}) &:= \mathrm{im}\left(R^c(\mathcal{M}_{g,n}) \xrightarrow{\mathrm{cl}} H^{2c}(\mathcal{M}_{g,n}; \mathbf{Q})\right), \\ \mathrm{RH}^{2c+1}(\mathcal{M}_{g,n}; \mathbf{Q}) &:= 0. \end{aligned} \quad (15)$$

A smooth algebraic stack X has the *Chow–Künneth generation property* if the exterior product

$$A^*(X) \otimes A^*(Y) \longrightarrow A^*(X \times Y)$$

is surjective for every finite-type stack Y admitting a stratification by global quotient stacks. Canning–Larson–Payne prove that if a smooth open substack of a smooth proper Deligne–Mumford stack has this property, then its cycle class map surjects onto all lowest-weight cohomology [6, Lemma 4.3].

Their Table 1 gives marking bounds

$$\begin{array}{c|cccccccc} g & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \hline c(g) & \infty & 10 & 10 & 11 & 11 & 7 & 5 & 3 \end{array} \quad (16)$$

such that, for $g \leq 7$ and $n \leq c(g)$,

$$W_k H^k(\mathcal{M}_{g,n}; \mathbf{Q}) = \mathrm{RH}^k(\mathcal{M}_{g,n}; \mathbf{Q}) \quad (17)$$

for every k [6, Proposition 4.5]. The geometric results behind most of these bounds come from Canning–Larson [5]; the direct locators used here are Table 1, Lemma 4.3, and Proposition 4.5 of [6].

The tautological vanishing needed below combines the pointed theorem of Ionel with the unpointed theorem of Looijenga.

Proposition 3.3 (Ionel–Looijenga). *For $g \geq 2$, tautological cohomology on the open moduli stack satisfies*

$$\mathrm{RH}^{2c}(\mathcal{M}_{g,n}; \mathbf{Q}) = 0 \quad \text{if} \quad \begin{cases} c \geq g, & n > 0, \\ c \geq g - 1, & n = 0. \end{cases} \quad (18)$$

Proof. For $n > 0$, Ionel proves that every product of total degree at least g in descendant and tautological classes vanishes in the rational cohomology of $\mathcal{M}_{g,n}$ [10, Theorem 0.1]. Since $R^*(\mathcal{M}_{g,n})$ is generated by the ψ - and κ -classes, every class in $R^c(\mathcal{M}_{g,n})$ with $c \geq g$ is a combination of such products, so its cohomology class vanishes. For $n = 0$, Looijenga proves that the rational tautological Chow ring of $\mathcal{M}_g$ vanishes in codimension greater than $g - 2$ [14, Theorem 1.2]. Passing through the cycle map and using (15) gives (18). $\square$

3.3. The critical line. Put $d = 3g - 3 + n$ and $k = 2d - 16$. For $2g + n = 18$, the number of markings $n = 18 - 2g$ is positive in every genus considered here, and

$$k = 4g - 4 + n = \mathrm{vcd}(\mathrm{PMod}_{g,n}). \quad (19)$$

The relevant cases in genera three through seven are displayed below.

TABLE 1. The critical open cases in genus at most seven.

(g, n)	d	$k = 2d - 16$	source of the open calculation
(3, 12)	18	20	$n - 1 = c(3)$; one-marking argument
(4, 10)	19	22	[6, Proposition 4.5]
(5, 8)	20	24	$n - 1 = c(5)$; one-marking argument
(6, 6)	21	26	$n - 1 = c(6)$; one-marking argument
(7, 4)	22	28	$n - 1 = c(7)$; one-marking argument

Proposition 3.4. *For*

$$(g, n, k) = (3, 12, 20), (5, 8, 24), (6, 6, 26), (7, 4, 28),$$

one has

$$W_k H^k(\mathcal{M}_{g,n}; \mathbf{Q}) = 0. \quad (20)$$

Proof. Fix one of the four triples and apply Proposition 3.1. Since $k = 4g - 4 + n$, the degree on the right of (13) is

$$k - n = 4g - 4 > 4g - 5 = \text{vcd}(\text{Mod}_g).$$

Lemma 3.2 makes every local-system summand vanish. Hence

$$W_k H^k(\mathcal{M}_{g,n}; \mathbf{Q}) = \Phi_{g,n}^k + \Psi_{g,n}^k. \quad (21)$$

Moreover,

$$\text{vcd}(\text{PMod}_{g,n-1}) = 4g - 4 + (n - 1) = k - 1,$$

so $H^k(\mathcal{M}_{g,n-1}; \mathbf{Q}) = 0$. Thus $\Phi_{g,n}^k = 0$.

It remains to consider the source of $\Psi_{g,n}^k$. In all four cases, $n - 1 = c(g)$. By [6, Table 1, Lemma 4.3, and Proposition 4.5],

$$W_{k-2} H^{k-2}(\mathcal{M}_{g,n-1}; \mathbf{Q}) = \text{RH}^{k-2}(\mathcal{M}_{g,n-1}; \mathbf{Q}).$$

The four Chow codimensions $(k - 2)/2$ are respectively 9, 11, 12, and 13, each at least g . Proposition 3.3 therefore makes these source groups zero. Equations (12) and (21) now imply (20). $\square$

Corollary 3.5. *All five groups in (4) vanish. Equivalently, for $3 \leq g \leq 7$ and $2g + n = 18$,*

$$W_{-16} H_{16}^{\text{BM}}(\mathcal{M}_{g,n}; \mathbf{Q}) = 0. \quad (22)$$

Proof. Proposition 3.4 handles (3, 12), (5, 8), (6, 6), and (7, 4). At (4, 10), Proposition 4.5 of [6] identifies the pure weight group with tautological cohomology; its codimension is $11 \geq 4$, so Proposition 3.3 makes it zero.

For every row, $k = 2d - 16$. Poincaré duality (5) gives

$$W_{-16} H_{16}^{\text{BM}}(\mathcal{M}_{g,n}; \mathbf{Q}) \cong W_k H^k(\mathcal{M}_{g,n}; \mathbf{Q})(d),$$

which proves (22). $\square$

Remark 3.6. A recent preprint of Liu [13] announces the stronger Chow-theoretic statements that $A^*(\mathcal{M}_{5,8})$ and $A^*(\mathcal{M}_{5,9})$ are tautological and that these stacks have the Chow–Künneth generation property, by a direct analysis of configurations of points on canonical genus-five curves. Those results would give the (5, 8) row of (4) by the same mechanism as the (4, 10) row. The argument given here is independent of that preprint: the (5, 8) case of Proposition 3.4 uses only the published range $c(5) = 7$ of [6, 5], together with Proposition 3.1, Harer’s theorem, and Ionel’s theorem.

3.4. The open term throughout the range.

Proposition 3.7. *Let $3 \leq g \leq 7$ and let (g, n) be stable. Then, for some $a_{g,n} \geq 0$,*

$$W_{-16}H_{16}^{\text{BM}}(\mathcal{M}_{g,n}; \mathbf{Q}) \cong \mathbf{Q}(8)^{\oplus a_{g,n}}. \quad (23)$$

Moreover, $a_{g,n} = 0$ whenever $2g + n \geq 11$.

Proof. If $n \geq 2$ and $2g + n > 18$, the Borel–Moore group itself vanishes by [2, Proposition 2.1]; indeed, that proposition gives

$$H_i^{\text{BM}}(\mathcal{M}_{g,n}; \mathbf{Q}) = 0 \quad (i < 2g - 2 + n).$$

We may therefore assume either $n \leq 1$ or $2g + n \leq 18$.

Put $d = 3g - 3 + n$ and $j = 2d - 16$. If $j < 0$, the group is zero by (5). Assume $j \geq 0$. The remaining marking bounds are

g	3	4	5	6	7
largest n	12	10	8	6	4.

For $g = 3, 5, 6, 7$, every pair with n strictly below the listed bound lies in the ranges (16), and for $g = 4$ every pair through $(4, 10)$ does. The only pairs outside these direct ranges are $(3, 12)$, $(5, 8)$, $(6, 6)$, and $(7, 4)$.

In the direct Canning–Larson–Payne range, (17) shows that $W_j H^j$ is algebraic and hence a direct sum of $\mathbf{Q}(-j/2)$: here j is even, and every codimension- $j/2$ cycle class has Hodge type $(j/2, j/2)$. The four exceptional groups vanish by Proposition 3.4. Therefore, in every remaining case,

$$W_j H^j(\mathcal{M}_{g,n}; \mathbf{Q}) \cong \mathbf{Q}(-j/2)^{\oplus a_{g,n}}$$

for some $a_{g,n} \geq 0$. Since $j/2 = d - 8$, Poincaré duality gives

$$\begin{aligned} W_{-16}H_{16}^{\text{BM}}(\mathcal{M}_{g,n}; \mathbf{Q}) &\cong W_j H^j(\mathcal{M}_{g,n}; \mathbf{Q})(d) \\ &\cong \mathbf{Q}(-j/2)(d)^{\oplus a_{g,n}} = \mathbf{Q}(8)^{\oplus a_{g,n}}. \end{aligned}$$

Suppose finally that $2g + n \geq 11$. If $n > 0$, then

$$\frac{j}{2} = 3g + n - 11 \geq g,$$

so Proposition 3.3 makes every direct-range group above zero. If $n = 0$, the inequality forces $g \geq 6$, and $j/2 = 3g - 11 \geq g - 1$, so the unpointed clause of the same proposition applies. The four exceptional groups and the cases removed at the start of the proof already vanish. Hence $a_{g,n} = 0$ in the asserted range. $\square$

4. COMPACT LOW-DEGREE INPUTS AND PRODUCTS

We collect the compact results used in the boundary induction.

Proposition 4.1. *The following statements hold for stable pairs.*

- (i) $H^q(\overline{\mathcal{M}}_{g,n}; \mathbf{Q}) = 0$ for odd $q \leq 9$.
- (ii) If $0 \leq q \leq 14$ is even, then

$$H^q(\overline{\mathcal{M}}_{g,n}; \mathbf{Q}) \cong \mathbf{Q}(-q/2)^{\oplus b_q}$$

for some $b_q \geq 0$.

- (iii) For every n , the group $H^{16}(\overline{\mathcal{M}}_{0,n}; \mathbf{Q})$ is a direct sum of copies of $\mathbf{Q}(-8)$ whenever $(0, n)$ is stable.
- (iv) For every stable $(1, n)$ and $(2, n)$,

$$H^{16}(\overline{\mathcal{M}}_{g,n}; \mathbf{Q}) \cong \mathbf{Q}(-8)^{\oplus b_{g,n}}.$$

Proof. Part (i) is the theorem of Bergström–Faber–Payne [2, Theorem 1.1]. For part (ii), recall that Canning–Larson–Payne write

$$H_q(\overline{\mathcal{M}}_{g,n}) := H^{2d-q}(\overline{\mathcal{M}}_{g,n}), \quad d = 3g - 3 + n.$$

Their Theorem 1.5(3) says that this complementary cohomology is tautological for even $q \leq 14$. Hence it is a sum of copies of $\mathbf{Q}(-(d-q/2))$. Poincaré duality in the form

$$H^q(\overline{\mathcal{M}}_{g,n}) \cong H^{2d-q}(\overline{\mathcal{M}}_{g,n})^\vee(-d)$$

gives the Tate type $\mathbf{Q}(-q/2)$. This transfers Hodge type, not tautological generation in degree q .

Keel proves that the cycle class map from the Chow ring of $\overline{\mathcal{M}}_{0,n}$ to cohomology is an isomorphism [11], which gives part (iii). Petersen proves that the even cohomology of $\overline{\mathcal{M}}_{1,n}$ is spanned by algebraic strata classes and is therefore tautological [16, Corollary 1.2]. Finally, [6, Corollary 2.7] states directly that

$$H^{2k}(\overline{\mathcal{M}}_{2,n}; \mathbf{Q}) = \mathrm{RH}^{2k}(\overline{\mathcal{M}}_{2,n}; \mathbf{Q}) \quad (k \leq 10).$$

Taking $k = 8$ proves part (iv). □

Lemma 4.2. *Let (g_1, n_1) and (g_2, n_2) be stable. If $H^{16}(\overline{\mathcal{M}}_{g_i, n_i}; \mathbf{Q})$ is a direct sum of copies of $\mathbf{Q}(-8)$ for $i = 1, 2$, then*

$$H^{16}(\overline{\mathcal{M}}_{g_1, n_1} \times \overline{\mathcal{M}}_{g_2, n_2}; \mathbf{Q}) \cong \mathbf{Q}(-8)^{\oplus N} \quad (24)$$

for some $N \geq 0$.

Proof. The Künneth isomorphism is compatible with Hodge structures:

$$H^{16}(X \times Y; \mathbf{Q}) \cong \bigoplus_{p+q=16} H^p(X; \mathbf{Q}) \otimes H^q(Y; \mathbf{Q}).$$

If p and q are odd, then $\min(p, q) \leq 7$, so one factor vanishes by Proposition 4.1(i). If $p, q > 0$ are even, then both are at most fourteen, and part (ii) identifies their tensor product with a sum of

$$\mathbf{Q}(-p/2) \otimes \mathbf{Q}(-q/2) = \mathbf{Q}(-8).$$

The two remaining summands have bidegrees $(16, 0)$ and $(0, 16)$ and are covered by the hypotheses, since every moduli stack in question is connected and has $H^0 \cong \mathbf{Q}(0)$. $\square$

5. PROOF IN GENUS AT MOST SEVEN

Proof of Theorem 1.1. We argue by induction on $d = 3g - 3 + n$, simultaneously for all stable pairs with $g \leq 7$. The cases $g = 0, 1, 2$ hold for every n by Proposition 4.1(iii)–(iv), so they may be used at any stage of the induction. Assume $3 \leq g \leq 7$.

Proposition 3.7 gives

$$W_{-16}H_{16}^{\text{BM}}(\mathcal{M}_{g,n}; \mathbf{Q}) \cong \mathbf{Q}(8)^{\oplus a_{g,n}}. \quad (25)$$

Consider a one-edge stable graph Γ of type (g, n) . Every vertex has genus at most g , and partial normalization at the distinguished node lowers the total complex dimension by one. More explicitly, a loop has ordered atlas $\overline{\mathcal{M}}_{g-1, n+2}$ of dimension $d - 1$, while a separating edge has ordered atlas

$$\overline{\mathcal{M}}_{g_1, n_1} \times \overline{\mathcal{M}}_{g_2, n_2}, \quad \dim \overline{\mathcal{M}}_{g_1, n_1} + \dim \overline{\mathcal{M}}_{g_2, n_2} = d - 1.$$

Each individual factor has dimension smaller than d .

For a loop, the induction hypothesis applies to the single factor. For a separating edge, it applies to both factors, and Lemma 4.2 gives the required type for their product. Taking the finite graph-automorphism invariants in (8) preserves this type. Summing over the finitely many one-edge graphs in (7), we obtain

$$H^{16}(\partial \widetilde{\mathcal{M}}_{g,n}; \mathbf{Q}) \cong \mathbf{Q}(-8)^{\oplus N_{g,n}}, \quad (26)$$

$$H_{16}(\partial \widetilde{\mathcal{M}}_{g,n}; \mathbf{Q}) \cong \mathbf{Q}(8)^{\oplus N_{g,n}}. \quad (27)$$

The boundary image $I_{g,n}$ in (10) is a quotient of (27). Hence $I_{g,n} \cong \mathbf{Q}(8)^{\oplus e_{g,n}}$ for some $e_{g,n}$. Combining this with (25), the boundary sequence becomes

$$0 \longrightarrow \mathbf{Q}(8)^{\oplus e_{g,n}} \longrightarrow H_{16}(\overline{\mathcal{M}}_{g,n}; \mathbf{Q}) \longrightarrow \mathbf{Q}(8)^{\oplus a_{g,n}} \longrightarrow 0. \quad (28)$$

The middle term is the dual of a polarizable pure Hodge structure of weight 16, so it is polarizable and pure of weight -16 . Semisimplicity splits (28) and gives

$$H_{16}(\overline{\mathcal{M}}_{g,n}; \mathbf{Q}) \cong \mathbf{Q}(8)^{\oplus (a_{g,n} + e_{g,n})}.$$

Dualizing proves (1). Every use of the induction hypothesis was at strictly smaller dimension, so the induction is well founded. $\square$

6. THE SIX-POINTED GENUS-SIX CASE AND THE GENUS-EIGHT REDUCTION

6.1. Boundary surjectivity in the genus-six case. The dimension calculation is

$$\dim_{\mathbf{C}} \overline{\mathcal{M}}_{6,6} = 3 \cdot 6 - 3 + 6 = 21.$$

The nonseparating component of its normalized boundary is

$$[\overline{\mathcal{M}}_{5,8}/S_2], \quad (29)$$

where S_2 exchanges the two branches of the distinguished node. A separating component is a finite quotient of

$$\overline{\mathcal{M}}_{g_1, n_1} \times \overline{\mathcal{M}}_{g_2, n_2}, \quad g_1 + g_2 = 6, \quad n_1 + n_2 = 8. \quad (30)$$

All vertex genera are at most six. Theorem 1.1, Lemma 4.2, and rational invariants therefore show that, for some $N \geq 0$,

$$H^{16}(\partial \widetilde{\mathcal{M}}_{6,6}; \mathbf{Q}) \cong \mathbf{Q}(-8)^{\oplus N}, \quad H_{16}(\partial \widetilde{\mathcal{M}}_{6,6}; \mathbf{Q}) \cong \mathbf{Q}(8)^{\oplus N}. \quad (31)$$

Proof of Corollary 1.2. Corollary 3.5 gives

$$W_{-16} H_{16}^{\text{BM}}(\mathcal{M}_{6,6}; \mathbf{Q}) = 0.$$

The right-exact sequence (9) is therefore exactly the surjection (2). Its target is a quotient of the second group in (31), so

$$H_{16}(\overline{\mathcal{M}}_{6,6}; \mathbf{Q}) \cong \mathbf{Q}(8)^{\oplus b_{16}}.$$

Proper duality gives the asserted cohomological statement. $\square$

The conclusion concerns the image of boundary *homology*. It does not assert that $H^{16}(\overline{\mathcal{M}}_{6,6})$ is generated by algebraic boundary cycles.

6.2. Reduction to two genus-eight inputs. The Borel–Moore vanishing used in Proposition 3.7 holds in all genera:

$$H_{16}^{\text{BM}}(\mathcal{M}_{g,n}; \mathbf{Q}) = 0 \quad \text{if} \quad \begin{cases} 16 < 2g, & n = 0, 1, \\ 16 < 2g - 2 + n, & n \geq 2. \end{cases} \quad (32)$$

This is [2, Proposition 2.1] together with duality between Borel–Moore homology and compactly supported cohomology.

The equality case left open by (32) at $(g, n) = (8, 2)$ is settled by Wong:

$$H^{30}(\mathcal{M}_{8,2}; \mathbf{Q}) = 0, \quad H_{16}^{\text{BM}}(\mathcal{M}_{8,2}; \mathbf{Q}) = 0. \quad (33)$$

Indeed, the first equality is the specialization to $g = 8$ of [18, Theorem 4.1], which states $H^{4g-2}(\mathcal{M}_{g,2}; \mathbf{Q}) = 0$ for every $g \geq 1$. Since $\dim_{\mathbf{C}} \mathcal{M}_{8,2} = 23$, the second equality follows from (5).

Proof of Theorem 1.3. By Theorem 1.1, all vertex types of genus at most seven are already known. We first use the two assumed open groups to establish the first compact genus-eight base cases. For $(8, 0)$, every vertex of a one-edge stable graph has genus at most seven: a genus-eight vertex on a separating edge would leave a vertex of type $(0, 1)$, which is unstable, and the loop atlas is $\overline{\mathcal{M}}_{7,2}$. The boundary argument from Section 5, together with the assumed open quotient, proves the compact assertion. The same is true for $(8, 1)$: a rational component carrying the unique original marking and one branch of the node would be unstable, so no genus-eight vertex occurs in its normalized boundary.

For $(8, 2)$, the open term vanishes by (33). The only new possibility in its normalized boundary with a genus-eight vertex is a rational tail carrying the two original markings; the other factor is then of type $(8, 1)$, which has just been established. The boundary sequence therefore proves the compact assertion for $(8, 2)$ without a third genus-eight hypothesis.

We now prove the assertion for all remaining stable pairs by induction on $d = 3g - 3 + n$.

For $g = 8$ and $n \geq 3$, the second inequality in (32) holds, while $n = 2$ is covered by (33). For $g \geq 9$, it holds for every $n \geq 2$, and the first inequality holds for $n = 0, 1$. Thus the open quotient in (10) is zero outside the two assumed genus-eight pairs.

Every vertex of a one-edge graph has smaller dimension than the original pair. The induction hypothesis, Lemma 4.2, and finite-group invariants therefore give Tate type $\mathbf{Q}(-8)$ for the degree-sixteen cohomology of the normalized boundary. Its homology is of type $\mathbf{Q}(8)$, so the boundary sequence and semisimplicity prove the assertion for the original pair. This completes the induction. $\square$

For clarity, the two remaining open groups correspond under (5) to the following ordinary lowest-weight groups:

n	$d = 21 + n$	$j = 2d - 16$	$j/2$	
0	21	26	13	
1	22	28	14	(34)

If $\mathcal{M}_{8,n}$ had the Chow–Künneth generation property and tautological rational Chow ring for $n = 0, 1$, Lemma 4.3 of [6] would identify $W_j H^j$ with tautological cohomology. Proposition 3.3 would then make both groups zero: the codimensions are 13 and 14, whereas the respective

vanishing thresholds are 7 and 8. Canning and Larson have proved that the rational Chow ring of $\mathcal{M}_8$ is tautological [4]; the additional Chow–Künneth input is a separate requirement, since tautologicality of the Chow ring alone does not control lowest-weight cohomology.

Remark 6.1. All results are over $\mathbf{C}$, use rational coefficients, and concern Deligne–Mumford stacks rather than their coarse moduli spaces. No integral, ℓ -adic, point-counting, or exact-rank statement is asserted.

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